

B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

IT 19-204-0304 DATABASE MANAGEMENT SYSTEMS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Explain database architecture and representation models.

CO2: Apply DDL and DML commands using SQL to retrieve data from the given table.

CO3: Apply normalization techniques to design a database for a given application.

CO4: Apply data storage techniques for a given scenario.

CO5: Describe concurrency control and transaction processing techniques.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) What is the need of DBMS? Write any two applications of data base system.	3	L2	1	1,2,3, 12	
(b) Illustrate strong entity set and weak entity set.	3	L2	1	1,2,3, 12	
(c) Differentiate super key, candidate key and foreign key.	3	L2	2	1,2,3, 4,12	
(d) Flights(<i>fno</i> : integer, <i>from</i> : string, <i>to</i> : string, <i>distance</i> : integer, <i>departs</i> : time, <i>arrives</i> : time) Aircraft(<i>aid</i> : integer, <i>aname</i> : string, <i>cruisingrange</i> : integer) Certified(<i>eid</i> : integer, <i>aid</i> : integer) Employees(<i>eid</i> : integer, <i>ename</i> : string, <i>salary</i> : integer) 1. Find the <i>eids</i> of pilots certified for some Boeing aircraft. 2. Find the <i>names</i> of pilots certified for some Boeing aircraft.	3	L3	1	1,2,3, 12	
(e) Consider R(A,B,C,D,E,H). Find the candidate key of the following functional dependency. A ->B BC->D E->C D->A	3	L3	2	1,2,3, 4,12	
(f) What are the ways of organizing records in a file?	3	L2	4	1,2,3, 12	
(g) How to recover from deadlock?	3	L2	5	1,2,3, 10,12	
(h) What is Thomas' write rule?	3	L1	5	1,2,3, 10,12	

(P.T.O.)

PART B

(4 × 12 = 48)

Marks BL CO PO

- | | | | | | | |
|-----|-----|--|---|----|---|--------------|
| II. | (a) | Describe the database architecture in detail. | 4 | L2 | 1 | 1,2,3,
12 |
| | (b) | A company has the following scenario: There are a set of salespersons. Some of them manage other salespersons. However, a salesperson cannot have more than one manager. A salesperson can be an agent for many customers. A customer is managed by exactly one salesperson. A customer can place any number of orders. An order can be placed by exactly one customer. Each order lists one or more items. An item may be listed in many orders. An item is assembled from different parts and parts can be common for many items. One or more employees assemble an item from parts. A supplier can supply different parts in certain quantities. A part may be supplied by different suppliers. Identify and list entities, suitable attributes, primary keys, and relationships to represent the scenario.

Draw an ER diagram to model the scenario . | 8 | L3 | 1 | 1,2,3,
12 |
- OR**
- | | | | | | | |
|------|-----|---|---|----|---|--------------|
| III. | (a) | Explain the following with example:
(i) Generalization (ii) Specialization (iii) Aggregation | 8 | L2 | 1 | 1,2,3,
12 |
| | (b) | Explain briefly different users in DBMS. | 4 | L2 | 1 | 1,2,3,
12 |
- | | | | | | | |
|-----|-----|--|---|----|---|----------------|
| IV. | (a) | Explain in detail about various aggregate functions used in SQL. Give example. | 8 | L2 | 2 | 1,2,3,
4,12 |
| | (b) | What is relational Algebra? Explain different operations of Relational Algebra with example. | 4 | L2 | 2 | 1,2,3,
4,12 |
- OR**
- | | | | | | | |
|----|-----|---|---|----|---|----------------|
| V. | (a) | Illustrate and explain different types of joins in SQL. | 8 | L2 | 2 | 1,2,3,
4,12 |
| | (b) | Consider the following database schema for supplier-parts-projects database:

Supplier(Sno,Sname,date_of_Birth,birth_place)
Parts(Pno,Pname,color, weight,city).
Project(Jno,Jname,city)
Shipment(Sno,Pno,Jno,qty)
Write queries in SQL to :

(i) Create all tables and set primary key and foreign key constraints.
(ii) Retrieve the average quantity supplied to each project.
(iii) Add a new constraint on Parts weight : Weight should not be null.
(iv) Display the total quantity supplied by supplier 'Hari'. | 4 | L3 | 2 | 1,2,3,
4,12 |

(Continued)

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		Marks	BL	CO	PO
-VI.	(a) What is normalization? Why do we need to normalize our database? Explain various normal forms with examples. Consider the relation R(a,b,c,d) with Set F={a→b,b→c}. Decompose this relation from 1 NF to 2 NF.	8	L3	3	1,2,3,12
	(b) Briefly explain Armstrong's Axioms in Functional Dependency.	4	L2	3	1,2,3,12
OR					
VII.	(a) What is Functional Dependency? Explain the procedure of calculating the Canonical Cover of a given Functional Dependency Set with suitable example.	8	L2	3	1,2,3,12
	(b) Define hashing. Which are the different methods of hashing?	4	L1	3	1,2,3,12
VIII.	(a) Write a short note on conflict Serializability and View Serializability.	8	L2	5	1,2,3,10,12
	(b) Define transaction. Explain ACID properties of a transaction.	4	L1	5	1,2,3,10,12
OR					
IX.	(a) Describe two-phase locking protocol and graph based protocol.	8	L2	5	1,2,3,10,12
	(b) Explain with a state diagram which are the states of a transaction.	4	L2	5	1,2,3,10,12

Bloom's Taxonomy Levels

L1 - 9%, L2 - 69%, L3 - 22%.
